ECE - 510 SPRING 2026

CODEFEST-1

CMAN

TASKS:

1. To calculate the MACs for each layer

MACs (Total weight) = No. of Inputs x No. of Outputs

To calculate for each layer of 784 → 256 → 128 → 10

	LAYER-1 (784 → 256)	LAYER-2 (256 → 128)	LAYER-3 (128 → 10)
No. of Inputs	784	256	128
No. of Outputs	256	128	10
MACs = No. of Inputs x No. of Outputs	784 x 256 ⇒ 200,704 MACs	256 x 128 ⇒ 32,768 MACs	128 x 10 ⇒ 1280 MACs

2. Total MACs for one-forward pass (combining all three layers)

⇒ 200704 MACs + 32768 MACs + 1280 MACs

⇒ 234,752 MACs

3. The total no. of trainable parameters (no biases)

⇒ (784 x 256) + (256 x 128) + (128 x 10)

⇒ 200,704 + 32768 + 1280

⇒ 234,752 parameters.

4. The total weight memory in bytes (FP32 → 4 Bytes each)

$\Rightarrow 234,752 \times 4$

$\Rightarrow \underline{\underline{939,008}}$ Bytes

5. Total Activation Memory in Bytes

$\Rightarrow 784 + 256 + 128 + 10$

$\Rightarrow 1,178$

Each having one memory for each neuron

Total Activation Memory $\Rightarrow 1,178 \times 4$

$\Rightarrow \underline{\underline{4,712}}$ Bytes

6. Arithmatic Intensity $\Rightarrow \dfrac{2 \times \text{Total MACs}}{\text{weight Bytes} + \text{Activation Bytes}}$

$\Rightarrow \dfrac{2 \times 234,752}{939,008 + 4,712}$

$\Rightarrow \dfrac{469,504}{943,720}$

$\Rightarrow \underline{\underline{0.4975}}$ FLOP / Byte